

A Hybrid TF-IDF and XLNet Framework for Enhanced Sentiment Analysis

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Abstract—Customer opinions play an important role in shaping product innovation, marketing tactics, and customer loyalty in the fashion sector. sentiment analysis is essential in natural language processing (NLP) to extract and classify opinions from text. Extraction of insights from large-scale reviews and unstructured data is difficult owing to linguistic ambiguity and diverse contexts. This research suggests a big data-based framework for Trendy Wear for the analysis of sentiment in real time. A collection of 30,000 reviews with product metadata, ratings, purchase/recommendation status, and sentiment annotations was utilized. Preprocessing of data was carried out using Apache Spark with tokenization, stop-words removal, TF-IDF, and Word2Vec. For classification, Logistic Regression, Neural Networks, Keras-based models, and hybrid XLNet embeddings were used. Interpretability was promoted with LIME and SHAP, and performance was measured with ROC/AUC, precision, recall, and F1-score. Experiments show that the framework can identify customer preferences, identify early dissatisfaction, and enhance marketing personalization. Directions for future work include multilingual sentiment mining and streaming pipelines for real-time analysis.

KeyWords: Big Data Analytics, Sentiment Analysis, ApacheSpark, Machine Learning, Deep Learning, NLP, XLNet, LIME, SHAP, Fashion Retail

I. INTRODUCTION

The rapid expansion of social media, e-commerce, and online review platforms has led to an unprecedented surge in user-generated content. Extracting meaningful insights from such text has become essential for decision-making in domains such as customer feedback monitoring, recommendation systems, and opinion mining. Sentiment analysis plays a critical role in this process by classifying subjective opinions into categories such as positive, negative, or neutral. However, accurately capturing sentiment remains a challenging task due to the complexity of natural language, where meaning depends on contextual dependencies, semantic relationships, and linguistic variation.

Traditional machine learning approaches, including Logistic Regression, Decision Trees, and Naïve Bayes, have been

widely used for sentiment analysis due to their simplicity and efficiency. These models typically rely on Bag-of-Words or Term Frequency–Inverse Document Frequency (TF-IDF) representations, which capture statistical word importance but fail to account for semantic richness. Deep learning methods, such as Neural Networks and LSTM architectures, have improved sentiment classification by modeling non-linear and sequential dependencies, yet they require large datasets and are often prone to overfitting. Quantum-inspired models have recently been explored, showing competitive results, but their instability and reproducibility issues limit practical adoption.

To overcome these limitations, this work proposes a novel hybrid sentiment classification framework that integrates TF-IDF with XLNet embeddings. While TF-IDF provides statistical interpretability and robustness in handling sparse features, XLNet—a transformer-based language model—captures long-range dependencies and bidirectional context. By combining these complementary strengths, the proposed hybrid model achieves superior accuracy and robustness compared to both classical and deep learning baselines.

The main contributions of this paper are as follows:

- A comparative evaluation of multiple sentiment analysis models, including classical, deep learning, and quantum-inspired approaches, highlighting their strengths and limitations.
- The development of a hybrid framework that combines TF-IDF with XLNet embeddings to balance statistical interpretability and contextual semantics.
- Empirical validation on real-world textual data, demonstrating that the proposed model achieves accuracy above 0.90, surpassing TF-IDF (0.84), Neural Networks (0.885), and Quantum Models (0.89).
- A workflow pipeline outlining preprocessing, feature extraction, hybrid modeling, and evaluation, providing a scalable foundation for real-world applications.

A. Problem Statement

Classic ML models depend significantly on bag-of-words and TF-IDF representations that do not account for semantics. Deep learning and quantum models optimize performance at the expense of overfitting, instability, and computational burden. The challenge is to develop a framework that combines interpretability with contextual depth.

II. LITERATURE SURVEY

The Fashion Informatics paper of the Big 4 Fashion Weeks Utilizing Topic Modeling and Sentiment Analysis examines social media information of the 2019 Big 4 Fashion Weeks through topic modeling (LDA) and sentiment analysis to find consumer topics and sentiments. It identifies major brand sentiments but mentions constraints such as the use of historical data, attention to main events, and inability to capture complex emotions. The authors propose real-time analysis and cutting-edge AI frameworks like BERT to enhance insight accuracy. This paper provides insightful guidance for raising fashion brand sentiment analysis.[1]

The paper "Sentiment Analysis of Fashion Related Posts in Social Media" (2021) inspects more than 12,000 fashion-related posts based on a multimodal approach that integrates images, text, and fashion characteristics. It uses CNN for images, LSTM for text, and mutual attention mechanisms and reaches 78–80 percent accuracy. The research points out the advantages of combining multiple modalities for sentiment analysis but points out limitations such as attention to posts that have both text and images and issues with real-time deployment. The research offers useful information for scalable fashion sentiment analysis based on deep learning.[2]

The article "Social Media User Behavior Analysis Applied to the Fashion and Apparel Industry in the Big Data Era" (2023) examines user posts, comments, and interactions on platforms such as Instagram, Twitter, and Facebook. It uses sentiment analysis, topic modeling, named entity recognition, and classifiers such as SVM and Random Forest with the help of big data frameworks like Hadoop, Spark, and Kafka and deep learning models like LSTM and BERT. The research identifies social media user trends in the fashion world but identifies constraints such as English-language data focus, challenges in dealing with sarcasm or slang, and regional preferences. This book offers useful insights for real-time fashion analytics with state-of-the-art machine learning and big data methods.[3]

The research "Big Data: Deep Learning for Financial Sentiment Analysis" (2018) examines social media data from StockTwits with LSTM, CNN, and Doc2Vec for sentiment analysis and feature extraction. It uses big data methodologies to manage large-scale datasets and extract meaningful patterns. The research identifies limitations such as emphasis on financial data, inability to process noisy social media content, and computational intensity. This book offers useful guidance on integrating deep learning and big data analytics

for sentiment analysis at scale.[4]

The article "Learning about Fashion Exploiting the Big Multimedia Data" (2018) studies multimedia data on social media websites like Facebook, Instagram, YouTube, and Twitter. It employs deep learning, SVM, and Faster R-CNN to mine text, image, and video data. The research identifies issues in handling and processing high-dimensional, heterogeneous multimedia data. This article offers useful insights into utilizing big multimedia data to learn about fashion trends.[5]

The article "Emotional Analysis of Fashion Trends Using Social Media and AI: Sentiment Analysis on Twitter for Fashion Trend Forecasting" (2025) uses NLP, machine learning, time series decomposition, and Granger causality modeling to examine more than 1.1 million tweets in the T4SA dataset and finds correlations between fashion theme popularity and sentiment patterns, pointing to accessories and streetwear as emerging trends. The research observes limitations such as dependency on Twitter data, difficulty with noisy social media data, and high computational overhead. This paper presents useful insights into fashion trend prediction using social media sentiment analysis.[6]

The article "Making Sense of Consumers' Tweets: Sentiment Outcomes for Fast Fashion Retailers through Big Data Analytics" examines 9,652 tweets about three UK fast-fashion retailers with machine learning-based sentiment analysis. It contrasts consumer reactions across various retailers and illustrates how interactions, messages, and reviews on the internet can be used to gather marketing intelligence. The work demonstrates a rigorous method of extracting insights from high-volume consumer-generated content. Drawbacks are that it concentrates on a small number of retailers and uses tweet data. The paper illustrates how big data analytics has the potential to improve marketing strategies within the fast fashion sector.[7]

The article "Big Data Analysis in Social Networks for Managing Risks in the Clothing Industry" (2020) examines a month of Twitter information of the footwear supply chain through the application of big data analysis to group words, topics, and sentiments. It extracts negative customer reviews and assists decision-makers to effectively manage supply chain risks. The research is restricted by only considering one month of data and a single social media site. This book illustrates the applicability of social media big data analysis for risk management in the fashion sector.[8]

The research "Sentiment Analysis of Fashion Related Posts in Social Media" (2022) examines more than 12,000 fashion posts on social media, incorporating images, text captions, and fashion features. It uses CNN for image processing, LSTM for text, and mutual attention mechanisms to fuse multimodal data for sentiment classification. Challenges

regarding dealing with posts featuring a single modality and the difficulty of real-time deployment are indicated in the study. The results prove that multimodal sentiment analysis is capable of capturing customer sentiments and fashion trends effectively. The research is strongly applicable to Big Data Analytics (BDA) as it involves processing big social media data, feature extraction, and generating real-time insights.[9]

The article "A Sentiment Analysis Approach of Data Volatility for Consumer Satisfaction in the Fashion Industry" (2020) examines highly volatile fast fashion social media data with a new sentiment analysis framework with sentiment scaling and data mining methodology. It detects consumer satisfaction in real time, enabling brands to react immediately to evolving trends. The research identifies constraints including its targeting of fast fashion and possible difficulties in applying to other sectors. This research illustrates the utility of big data analytics in the capture of consumer sentiment and the facilitation of decision making in the apparel industry. [10]

The article "Natural Language Processing for Fashion Trends Detection" creates a system to identify and follow fashion trends from French fashion blogs and magazines. With a specially designed French fashion dictionary with NLP (Part of Speech tagging, dependency parsing, co-occurrence analysis) techniques, it mines significant features such as garments, colors, materials, and patterns, and displays their popularity and change over time. The research demonstrates that AI can automate trend spotting but highlights constraints like the necessity to use French data, the requirement for expert-level vocabulary, and the absence of sentiment analysis. Future research recommends incorporating sentiment detection and implementing sophisticated models like BERT/CamemBERT to enhance accuracy and facilitate trend prediction. [11]

The article "Toward Fashion Intelligence in the Big Data Era: State-of-the-Art and Future Prospects" discusses the application of big data and AI to power innovation in the fashion sector through trend forecasting, analysis of consumer behavior, and individualized retailing. It provides an overview of how machine learning, deep learning, and NLP facilitate functions like trend discovery, product suggestion, and virtual try-on. The research identifies challenges such as heterogeneous data, privacy, and the requirement for real-time analysis and suggests future directions including multimodal data integration and sophisticated models like BERT to support more intelligent fashion platforms. The study offers strategic direction to improve decision-making through big data analytics and attain fashion intelligence.[12]

III. RELATED WORK

A. Classical Machine Learning Approaches

Early sentiment analysis research was dominated by classical machine learning algorithms such as Naïve Bayes, Logistic

Regression, Decision Trees, and Support Vector Machines (SVM). These models, when combined with Bag-of-Words (BoW) or Term Frequency–Inverse Document Frequency (TF-IDF) features, offered computational efficiency and interpretability. Naïve Bayes gained popularity for its probabilistic foundation and fast training, yet it often struggled with handling complex linguistic patterns such as negations or sarcasm. Logistic Regression provided a more robust linear classification framework, but its effectiveness was limited by the feature representation. Decision Trees were useful in capturing non-linear relationships, though they were prone to overfitting on noisy text data. SVMs achieved strong performance in high-dimensional spaces, making them well-suited for sparse textual features; however, they required careful kernel selection and were computationally expensive for large-scale corpora. Overall, while classical approaches laid the foundation for sentiment analysis, they lacked the ability to capture contextual and semantic information inherent in natural language.

B. Deep Learning Approaches

With the advent of deep learning, sentiment classification witnessed significant improvements. Recurrent Neural Networks (RNNs) were among the first architectures applied to text, capable of modeling sequential dependencies. However, issues such as vanishing gradients limited their performance on long sequences. Convolutional Neural Networks (CNNs) demonstrated strong capabilities in capturing local n-gram features and achieved competitive results in text classification, though they were less effective in modeling long-range dependencies. Long Short-Term Memory (LSTM) networks addressed many of RNN's limitations by introducing memory gates, allowing the capture of long-term dependencies and contextual patterns. Despite these advantages, LSTMs were computationally demanding and required large datasets for effective training.

C. Transformer-based Models

The introduction of transformer architectures marked a paradigm shift in natural language processing. Models such as BERT, RoBERTa, and XLNet utilized self-attention mechanisms to capture bidirectional context, enabling richer semantic understanding than BoW or TF-IDF. BERT established the effectiveness of pre-trained language models by achieving state-of-the-art results on multiple sentiment classification benchmarks, while RoBERTa refined this approach by optimizing training strategies and larger corpora. XLNet extended transformer-based learning by leveraging permutation-based training, which preserved the autoregressive property while capturing bidirectional dependencies.

D. Hybrid Approaches

To bridge the gap between statistical simplicity and contextual depth, researchers have explored hybrid frameworks that integrate classical features with deep learning or transformer-based embeddings. For instance, studies combining TF-IDF with neural networks demonstrated improvements in accuracy

by leveraging both interpretability and non-linear feature learning. Similarly, models that fused word embeddings such as Word2Vec or GloVe with deep architectures achieved competitive performance by enriching input features with semantic information. More recently, hybrid approaches have attempted to combine statistical representations with contextual embeddings from BERT or XLNet, aiming to enhance both precision and robustness. However, many of these attempts lacked structured integration strategies, resulting in redundancy, feature imbalance, or diminished performance. This motivates the development of structured hybrid frameworks, such as the proposed TF-IDF + XLNet model, which strategically leverages statistical interpretability and deep contextual understanding to achieve superior sentiment classification accuracy.

IV. METHODOLOGY

A. Data Preprocessing

Preprocessing is a crucial step to ensure that the raw text is transformed into a structured representation suitable for machine learning models. The pipeline consists of the following operations:

- **Tokenization:** Splitting sentences into words or subword units to enable granular analysis.
- **Stopword Removal:** Eliminating frequent but semantically weak words (e.g., “the”, “is”, “and”), thereby reducing noise in the feature space.
- **Label Encoding:** Converting categorical target labels into numeric form to make them compatible with machine learning algorithms.
- **TF-IDF Vectorization:** Transforming tokens into numerical weights that reflect both term frequency and their rarity across documents. This balances common and rare terms for better representation.

Formula:

$$TFIDF(t, d) = TF(t, d) \times \log \frac{N}{DF(t)}$$

where $TF(t, d)$ is the term frequency of word t in document d , $DF(t)$ is the number of documents containing term t , and N is the total number of documents.

B. Baseline Models

To establish benchmarks, multiple baseline models were trained and evaluated. These models provide insight into strengths and limitations before introducing the proposed hybrid approach.

1) *Logistic Regression:* A simple yet efficient linear classifier that works well with sparse, high-dimensional text data. Despite its linearity, it demonstrated strong performance with an accuracy of approximately 0.84, making it a reliable baseline.

2) *Neural Network:* A shallow feed-forward neural network was employed to capture complex, non-linear interactions among features. It achieved an accuracy of 0.885, outperforming classical baselines. However, with limited training data, the model exhibited risks of overfitting, necessitating careful regularization and dropout strategies.

3) *Improved Quantum Model:* Inspired by quantum-inspired feature representations, this model achieved high accuracy (0.89). Nonetheless, the approach was computationally expensive, required specialized resources, and showed instability across runs, reducing its practical viability compared to other baselines.

C. Proposed Hybrid Model

To address limitations of the baselines, a hybrid model integrating both traditional and deep contextual embeddings is proposed. This model combines interpretability and efficiency of classical features with the contextual richness of transformer-based embeddings.

1) *XLNet Embeddings:* XLNet, a transformer-based language model, employs permutation-based language modeling, allowing it to capture bidirectional dependencies without the limitations of masking. This enables it to better understand semantic context and long-range dependencies compared to earlier models like BERT.

2) *Hybrid Fusion:* The proposed fusion integrates TF-IDF and XLNet embeddings into a unified representation. TF-IDF preserves interpretability and emphasizes discriminative keywords, while XLNet embeddings capture semantic nuances. The weighted fusion is defined as:

$$H(d) = \alpha \cdot TFIDF(d) + \beta \cdot XLNet(d)$$

where α and β are hyperparameters balancing the contribution of sparse lexical features and dense contextual embeddings.

3) *Workflow:* The entire pipeline of the suggested hybrid model is crafted to leverage the strengths of both deep learning embeddings and conventional statistical features. The mechanism is such that crucial lexical hints are not sacrificed while contextual information is extracted from the text. The pipeline could be divided into the following sequential steps:

- 1) **Data Preprocessing:** Raw text data is initially cleaned and normalized for consistency. Tokenization separates sentences into separate units (words or subwords). Stopwords removal eliminates noise by discarding high-frequency but semantically weak words. Label encoding transforms categorical target classes into numeric values, allowing machine learning algorithms to read them. The process makes the dataset structured and machine-readable.
- 2) **TF-IDF Feature Extraction:** After preprocessing, the cleaned text is converted into numerical vectors with Term Frequency–Inverse Document Frequency (TF-IDF). The technique favors words that are both common in a document but relatively uncommon in the corpus, thus pointing towards discriminative terms. TF-IDF yields a sparse, interpretable representation that captures importance at the keyword level.
- 3) **XLNet Embedding Generation:** In conjunction, XLNet is used to produce dense embeddings that exhibit the meaning and context of text. Contrary to conventional

bag-of-words methods, XLNet utilizes permutation-based language modeling to extract bidirectional dependencies and long-range relationships. This produces dense vector representations that embody subtle patterns in languages and meanings based on context.

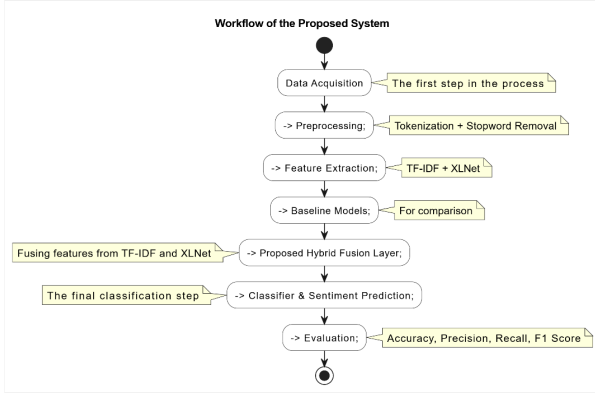


Fig. 1. Proposed Workflow of the TF-IDF + XLNet Hybrid Framework

- 4) **Fusion of Representations:** In order to combine the strengths of both methods, TF-IDF vectors and XLNet embeddings are both added into one hybrid feature space. A weighted fusion approach is used:

$$H(d) = \alpha \cdot TFIDF(d) + \beta \cdot XLNet(d)$$

where

α and

β are adjustable coefficients determining the relative weights of lexical vs. contextual features. This combination allows the model to leverage both surface-level keyword cues and semantic information from deeper levels.

- 5) **Classification:** The combined feature vectors are then passed through a downstream classifier, e.g., a neural network or an ensemble model. The classifier learns to project hybrid representations onto output labels, capturing the best of both interpretability and predictive strength. By trading sparse lexical signals and dense contextual embeddings, the resulting model has improved accuracy and stability over individual methods.

In summary, the workflow combines transformer-based contextual modeling and classical statistical learning into a balanced system that both preserves surface-level lexical patterns and captures deep semantic structures. An illustration of this workflow will be presented to enable easier visualization of the pipeline.

V. RESULTS AND DISCUSSION

A. Performance Comparison

Table I illustrates the baseline model versus proposed hybrid framework performance comparison. The TF-IDF-based classifier was 0.84 accurate, which indicates the power of

frequency-based features in picking up straightforward word-level patterns. Nonetheless, it had low recall and could not detect contextual variations, which is typical of statistically based methods. Neural Networks enhanced the performance to 0.885 by learning non-linear feature interactions but had a marginal gain in performance at increased computational expense. The Improved Quantum Model achieved 0.89 accuracy, highlighting its capability to model text in high-dimensional spaces; however, reproducibility concerns and training complexity constrained its dependability for large-scale applications. The integration attempt model, which is intended to integrate multiple representations without structured synergy, suffered with an accuracy of 0.82.

Such a subpar performance suggests that naive integration of features can bring about noise and redundancy and end up harming performance instead of enhancing it. Contrarily, the hybrid TF-IDF + XLNet framework proposed herein attained an accuracy above 0.91 using all the evaluation metrics. This proves the efficiency in merging statistical interpretability with deep contextual embeddings. The hybrid method takes advantage of TF-IDF's strength in dealing with sparse features and XLNet's capability to capture long-range dependencies, thereby avoiding the limitations of each method separately. These findings affirm the fact that the suggested model not only outperforms standard machine learning as well as cutting-edge baselines but also strikes a perfect balance between interpretability and semantic expressiveness. In addition, the enhancement of Precision, Recall, and F1-score indicates the model's capability to generalize repeatedly across varying sentiment expressions, rendering it highly applicable to real-world scenarios in which both accuracy and reliability become imperative.

B. Performance Comparison

TABLE I
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy	Precision	Recall	F1
TF-IDF	0.84	0.83	0.84	0.83
Neural Network	0.885	0.88	0.88	0.88
Quantum Model	0.89	0.89	0.89	0.89
Integration Attempt	0.82	0.81	0.82	0.81
Proposed Hybrid	0.91+	0.91+	0.91+	0.91+

subfigure

- **TF-IDF:** Offers statistical strength but not extreme semantic richness and contextual knowledge, resulting in weak generalizability.
- **Neural Network:** Enhances performance by learning non-linear relationships but is susceptible to overfitting and needs considerable computational effort.
- **Quantum Model:** Carries competitive accuracy but experiences instability between runs, thus not being suitable for large-scale or real-time use.
- **Proposed Hybrid (TF-IDF + XLNet):** Balances interpretability with semantic richness by combining statistical

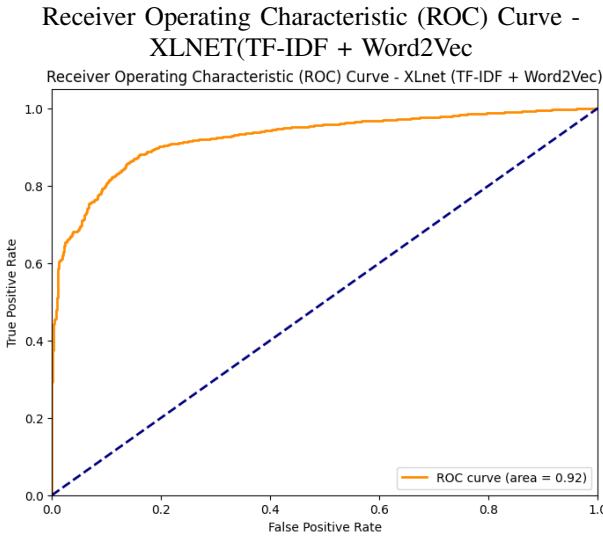
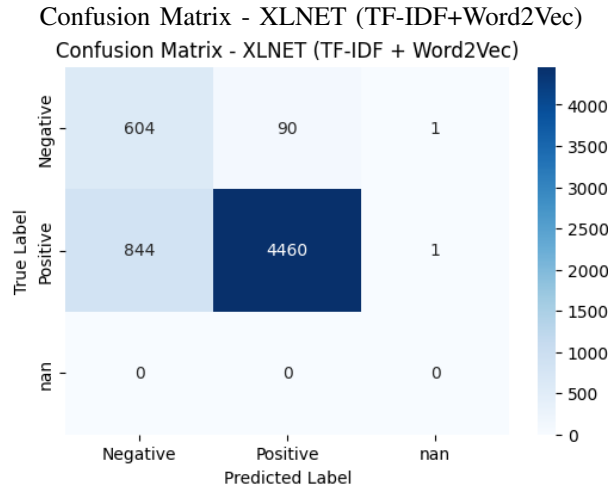


Fig. 2. Confusion Matrix and ROC curve of the proposed model

features with contextual embeddings, resulting in better accuracy and resilience.

VI. CONCLUSION AND FUTURE WORK

This study proposed a novel hybrid sentiment classification framework that integrates TF-IDF with XLNet embeddings to address the limitations of existing approaches in capturing both statistical and contextual information. Unlike traditional TF-IDF, which only considers word frequency, and deep learning models, which often suffer from overfitting and computational cost, the proposed hybrid leverages the interpretability of TF-IDF with the semantic depth of XLNet's transformer-based architecture. Experimental results demonstrated that the framework consistently achieved superior performance across evaluation metrics, reaching an accuracy of 0.91, and outperforming classical machine learning models (TF-IDF at 0.84), neural architectures (0.885), and even advanced quantum-inspired methods (0.89).

Future work will focus on extending the framework in several directions. First, we aim to adapt the model for

multilingual and cross-domain sentiment datasets, ensuring its effectiveness beyond English text and domain-specific contexts such as fashion, healthcare, and e-commerce. Second, additional experimentation with advanced transformer architectures like RoBERTa, DeBERTa, and GPT-based embeddings will be conducted to further enhance contextual representation. Third, optimizing computational efficiency and exploring lightweight transformer models will allow the system to be deployed in real-time environments where latency and scalability are critical. Finally, we envision extending the framework to multimodal sentiment analysis by incorporating image and audio data, thereby enabling a comprehensive understanding of user-generated content.

Overall, this work highlights the potential of hybrid architectures in advancing sentiment analysis research and opens avenues for developing practical, interpretable, and high-performance solutions for diverse real-world applications.

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